Positioning Embeddings, Attention and LightGBM: Methodology

PositioningWork project

1 November 2026

Status: a design only. Nothing here is built or tested. All sizes and settings are proposals. The design has no published precedent that we found. The evidence so far is weak: no positioning signal has passed an out-of-sample test in our studies. So every step must earn its place against a simpler model.

Summary

1. Turn positioning into features, keeping the full shape of the ledger, not one Pain number.
2. Test those features with ridge regression.
3. Test them with a small LightGBM (LGBM) model.
4. Add one small attention layer, freeze it, and pass its outputs to LGBM.
5. Keep a step only if it beats the step before it, out of sample.

Why attention at all. The literature suggests that a loss matters when it meets a binding limit, and that holder type can change the sign (Siriwardane 2019; Ellul et al. 2015; Kang et al. 2020; Fang 2024). Attention lets market state decide which reading to trust. LGBM can also learn such interactions, so attention may add nothing. Step 4 tests that.

1 Notation and data

A *unit* u is what we forecast, such as a bond, an issuer, a sector and maturity bucket, an ETF or an index. Time t is weekly. Every input must be known at t. Each feature carries the date it became available.

Market	Available positioning inputs	Limit on this method
TRACE	Customer buys and sells by bond	One holder group only, so attention can only weigh measures. Our pilot panel has 12 issuers and 123 weeks, and it is not point-in-time.
RFQ (work machine)	Anonymous side, size and level, from one dealer	No client view. Use buckets. Any signal must beat a vendor mark.
Credit ETFs	Shares, NAV, FINRA short interest	Share count is not a position. Prefer baskets and premiums if available.
Credit futures	CFTC holder groups	About 25 reports. Monitor only. Do not train.
CDX	Prices, volumes, skew, cleared totals	No public holder data. Proxies only.

2 Step 1: features

2.1 The ledger-shape embedding

This is the main new feature. For holder group g and unit u, build the lot ledger from net flow, with pro-rata removal. Entry level is the trade's spread, or total-return price for an ETF. For each open long lot ℓ, compute its distance from today, in volatility units:

$$D_\ell = \frac{s_t - s_\ell}{\sigma_t} \quad \text{for spreads,} \qquad D_\ell = \frac{P_\ell - P_t}{\sigma_t} \quad \text{for prices.}$$

A positive distance means the lot is under water. Short lots use the opposite sign. σ_t is the standard deviation of weekly changes over the past 26 weeks.

Place the lots in 16 bins, with edges from -3.5 to $+3.5$ in steps of 0.5, plus two open ends. Each bin holds that bin's share of open quantity:

$$h_b = \frac{\sum_{\ell \in b} q_\ell}{\sum_\ell q_\ell}, \qquad \sum_b h_b = 1.$$

Pain share and depth are summaries of $\mathbf{h}$. So $\mathbf{h}$ keeps everything they hold, and more.

Rules.

- If open quantity is 10^{-6} or less, mark the row missing. Add a missing flag.
- Always add coverage, which is open quantity over the amount outstanding or shares.
- Starting positions are unknown, so read $\mathbf{h}$ as the shape of recent flow, not of all holders.
- Run one test with Grinblatt and Han (2005) turnover decay, which also retires lots with volume. Our current ledger does not do this.

2.2 Other readings

Build these measures per group where the data allows.

Measure	Source in the spec
Net position scaled	I1, I3
Range position	I2 sentiment index, if history allows
Flow trend	TAARSS from I13
Filtered residual	I5
Directional short interest	ETFs only, following Evans et al. (2026)

Standardise each reading x with an expanding mean and standard deviation, known at t. Clip it at ± 3.

2.3 Market state

The state vector $\mathbf{s}_t$ has five inputs, all standardised: spread level, spread volatility, past 26-week spread change, a dealer capacity proxy, and a funding stress proxy. Choose and freeze the two proxies before any test.

2.4 Target

The target is the future spread change over H weeks, for $H \in \{1, 4, 13\}$, less the change in the unit's market bucket. Start with $H = 4$. Use spreads or excess returns, not price, because rates and pull-to-par distort price. Follow Dickerson et al. on bond data: avoid one-sided filters and allow for price error.

3 Step 2: ridge

Fit ridge regression on $\mathbf{h}$ for each group, the other readings, and $\mathbf{s}_t$. Choose the penalty inside the training data. This model tells you whether any linear signal exists. If it does not beat the benchmarks in Section 7, record that, and treat steps 3 and 4 as exploratory.

4 Step 3: LightGBM

Use the same inputs. Add hand-built interactions, such as depth $\times$ dealer capacity, and depth by holder group. This is the simple alternative to attention, and it has far less leakage risk.

Setting	Proposed value
Maximum depth, leaves	3, up to 8
Minimum rows per leaf	5% of training rows
Learning rate	0.02, with early stopping on a later block
Feature fraction	0.5
L2 penalty	10
Row weights	Each week sums to 1, so busy weeks do not dominate
Splits	By whole weeks, never by row

5 Step 4: attention

5.1 The layer

Each token i is one reading, with holder group g_i and measure m_i. Its tag, in $d = 8$ dimensions, is learned:

$$\mathbf{e}_i = \mathbf{e}^G_{g_i} + \mathbf{e}^M_{m_i}.$$

The query comes from market state:

$$\mathbf{q}_t = \mathbf{W}\mathbf{s}_t + \mathbf{c}.$$

The attention weights and the output are:

$$a_{i,t} = \frac{\exp(\mathbf{q}_t^\top \mathbf{e}_i/\tau)}{\sum_j \exp(\mathbf{q}_t^\top \mathbf{e}_j/\tau)}, \qquad z_{u,t} = \sum_i a_{i,t}\, b_{g_i,m_i}\, x_{i,u,t}.$$

The ledger shape enters as a single reading per group, $x = \mathbf{w}^\top \mathbf{h}$, with one vector $\mathbf{w}$ shared across groups. Set $\tau = 1$.

Why the signed loadings b. Weights a are always positive. Without b, the layer could not let insurers and levered holders carry opposite signs.

5.2 Training

A small linear head forecasts the target:

$$\hat{y}_{u,t} = \beta_0 + \beta z_{u,t} + \boldsymbol{\gamma}^\top \mathbf{s}_t.$$

Minimise Huber loss, plus an L2 penalty on all parameters, plus a small penalty for over-concentrated weights. Use Adam, whole-week batches and early stopping on a later block. Use three random seeds, and average the frozen outputs.

5.3 Size limit

With 3 groups, 5 measures, 16 bins and 5 state inputs, the layer has about 150 parameters:

Part	Parameters
Tags, $(3 + 5) \times 8$	64
Query, $5 \times 8 + 8$	48
Signed loadings, 3×5	15
Ledger-shape vector	16
Head, $1 + 1 + 5$	7
Total	150

Do not exceed this without new evidence. Use one head and one layer.

5.4 Passing it to LGBM

LGBM cannot train the layer, because trees pass no gradients back. So freeze the layer, then export these columns:

1. The weights $a_{i,t}$.
2. The output $z_{u,t}$.
3. The weight entropy $-\sum_i a_{i,t} \ln a_{i,t}$, which shows whether one reading dominates.

Add these to the inputs from Step 3. Do not export $\hat{y}$ as well, as it adds little beyond z and $\mathbf{s}_t$.

6 Preventing leakage

This is the main risk. If attention trains on the weeks LGBM learns from, its columns carry those weeks' targets, and LGBM will look far better than it is.

1. Set a final holdout before any fit. No model or choice touches it until the end.
2. Use an expanding walk-forward, refitting yearly. Everything fits only on data before the cut.
3. Inside each training window, split the weeks into five contiguous blocks.
4. For each block, fit attention on the other blocks, and write its columns for that block. Remove H weeks on each side of the block first. These out-of-fold columns are LGBM's training inputs.
5. Then fit attention on the whole training window. Use that fit to write the columns for test weeks.
6. Leave a gap of H weeks between training data and test data.
7. Compute all standardising, binning and volatility on past data only.
8. TRACE in our warehouse is not point-in-time. Label results from it as provisional.

7 Evaluation

7.1 The ladder

Model	Inputs	Question
B0	Spread level and past spread change	The bar to beat
M1	Ridge on all features	Is there linear signal?
M2	LGBM on the same features, plus hand-built interactions	Do interactions help?
M3	M2 plus frozen attention columns	Does attention add anything?

Each model must beat the one before it. The plain spread level had $t = 5.93$ on LQD and 12.51 on HYG at 26 weeks in our ETF study, so B0 is a demanding bar.

7.2 Statistics

- Primary measure: weekly cross-sectional rank correlation with the target, averaged over time. Use Newey-West errors with lag at least $H - 1$.
- Few issuers means weak clustering. Use a wild cluster bootstrap by issuer and date.
- Report effective independent observations for each horizon.
- Placebos: circularly shift flow within each unit, and shuffle holder labels. M3 must beat both.
- Apply a Holm correction across all models, horizons and markets. Report raw and adjusted p-values.
- Report a net result, after stated costs, for any trading claim.

7.3 Checks that attention means something

1. Weights should be stable across seeds and yearly refits. If they are not, do not read them.
2. Check whether weights shift sensibly in known stress, without using that check to tune the model.
3. Compare SHAP importance for attention columns against the placebo versions.

7.4 Stop rules

1. If M1 and M2 fail against B0, stop. Attention cannot create information that the inputs lack.
2. If M3 does not beat M2 out of sample, drop attention and keep the simpler model.
3. If weights are unstable, drop attention even if M3 scores slightly better.
4. Do not widen the search after seeing the holdout.

8 Order of work

1. The TRACE pilot passed 0 of 48 tests on 12 large issuers, and flow looked like noise. So do not start on that panel. Start only on a much larger one.
2. Build $\mathbf{h}$ on TRACE, then run B0 and M1.
3. If M1 passes, run M2.
4. If M2 passes, run M3 on a larger sample. Our current TRACE panel is too small to train attention reliably, with 123 weeks and 12 issuers. Treat any M3 result on it as exploratory.
5. Repeat the ladder on work-machine RFQ data, using buckets.

Sources

Full citations and reading levels are in `../literature/literature_review.pdf`. The ledger, measures and their recipes are in `../../positioning_spec.pdf`.